

Assignment 1.1.

Introduction to Data Cleaning

Objectives:

- Analyze and interpret the structure and content of raw datasets.
- Apply foundational data preprocessing techniques, such as handling missing values and data type conversions.
- Construct a basic data cleaning pipeline to process real-world data efficiently and with reproducibility.

Procedures

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

Download the dataset for this activity [here](#)

1. Load the dataset then show the first 5 and last 5 datapoints

```
import pandas as pd

df = pd.read_csv('/content/drive/MyDrive/VDA/dirty_cafe_sales.csv')
```

df.head()

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
0	TXN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway	2023-09-08
1	TXN_4977031	Cake	4	3.0	12.0	Cash	In-store	2023-05-16
2	TXN_4271903	Cookie	4	1.0	ERROR	Credit Card	In-store	2023-07-19
3	TXN_7034554	Salad	2	5.0	10.0	UNKNOWN	UNKNOWN	2023-04-27
4	TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet	In-store	2023-06-11

df.tail()

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
9995	TXN_7672686	Coffee	2	2.0	4.0	NaN	UNKNOWN	2023-08-30
9996	TXN_9659401	NaN	3	NaN	3.0	Digital Wallet	NaN	2023-06-02
9997	TXN_5255387	Coffee	4	2.0	8.0	Digital Wallet	NaN	2023-03-02
9998	TXN_7695629	Cookie	3	NaN	3.0	Digital Wallet	NaN	2023-12-02
9999	TXN_6170729	Sandwich	3	4.0	12.0	Cash	In-store	2023-11-07

QUESTION: What Column has numerical values?

The Columns that has numerical values are Quantity, Price per unit, Total Spent, and Transaction Date.

2. Check if there are any missing values in the dataset

```
df.describe()
```

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
count	10000	9667	9862	9821	9827	7421	6735	9841
unique	10000	10	7	8	19	5	4	367
top	TXN_9226047	Juice	5	3.0	6.0	Digital Wallet	Takeaway	UNKNOWN
freq	1	1171	2013	2429	979	2291	3022	159

df.isna().sum()

	0
Transaction ID	0
Item	333
Quantity	138
Price Per Unit	179
Total Spent	173
Payment Method	2579
Location	3265
Transaction Date	159

dtype: int64

QUESTION: How many columns had missing values? Which column had the highest number of missing values?

Seven Columns which are the Transaction Date, Item, Quantity, Price per Unit,Total Spent, Payment Method, Location, and Transaction Date. The Location has the highest number of missing values which has 3265 missing values.

3. Remove unnecessary columns and make sure numerical columns only contain numbers.

df.columns

Index(['Transaction ID', 'Item', 'Quantity', 'Price Per Unit', 'Total Spent', 'Payment Method', 'Location', 'Transaction Date'], dtype='object')

df = df.drop(columns=['Transaction ID', 'Payment Method', 'Location', 'Transaction Date'])

df.head()

	Item	Quantity	Price Per Unit	Total Spent
0	Coffee	2	2.0	4.0
1	Cake	4	3.0	12.0
2	Cookie	4	1.0	ERROR
3	Salad	2	5.0	10.0
4	Coffee	2	2.0	4.0

df.dtypes ## Checking the datatype for each column

	0
Item	object
Quantity	object
Price Per Unit	object
Total Spent	object

dtype: object

```
df["Quantity"] = pd.to_numeric(df["Quantity"], errors='coerce')
df["Price Per Unit"] = pd.to_numeric(df["Price Per Unit"], errors='coerce')
df["Total Spent"] = pd.to_numeric(df["Total Spent"], errors='coerce')

df.head()
```

	Item	Quantity	Price Per Unit	Total Spent
0	Coffee	2.0	2.0	4.0
1	Cake	4.0	3.0	12.0
2	Cookie	4.0	1.0	NaN
3	Salad	2.0	5.0	10.0
4	Coffee	2.0	2.0	4.0

```
df.dtypes ## Checking the datatype for each column
```

	dtype
Item	object
Quantity	float64
Price Per Unit	float64
Total Spent	float64

```
dtype: object
```

QUESTION: What happened to the object columns?

The object columns here were initially stored as raw text strings which includes also the non-numeric entries like the string 'ERROR' in the Total Spent column. When, the "pd.to_numeric(..., errors='coerce') function was applied to Quantity, Price Per Unit, and Total Spent, the Pandas parsed the valid numerical text into numbers and automatically converted the invalid 'ERROR' string into a missing NaN value because of a float64 data type

4. Remove missing data to ensure all that the final dataset is clean

```
df.isna().sum()
```

	sum
Item	333
Quantity	479
Price Per Unit	533
Total Spent	502

```
dtype: int64
```

```
df = df.dropna()
```

```
df.isna().sum()
```

	sum
Item	0
Quantity	0
Price Per Unit	0
Total Spent	0

```
dtype: int64
```

```
df.head()
```

	Item	Quantity	Price Per Unit	Total Spent
0	Coffee	2.0	2.0	4.0
1	Cake	4.0	3.0	12.0
3	Salad	2.0	5.0	10.0
4	Coffee	2.0	2.0	4.0
5	Smoothie	5.0	4.0	20.0

EXPLAIN the process of data cleaning done in the procedure.

The Data Cleaning Process began by identifying the missing values by the syntax `df.isna().sum()`, which revealed the significant amounts of missing or NaN entries across all the columns. To clean the dataset, `df = dropna()` was executed to remove any row containing at least one missing value. For the checking is the `df.isna().sum()` again to confirm the missing values across all columns were reduced to zero, while `df.head()` verified that incomplete rows such as index 2 were successfully removed.

Supplementary Activities

Download the dataset for supplementary activity [here](#)

1. Load the dataset and show the first 5 and last 5 datapoints
2. Check if there are any missing values in the dataset.
3. The df must only have Name, Subscribers, Category, and Country. Remove all other columns.
4. Convert Subscribers to type `int`
5. Ensure that there are no rows with missing values. (UNKNOWN is counted as a missing value)

```
import pandas as pd
```

```
df = pd.read_csv('/content/drive/MyDrive/VDA/youtube_subscribers_data.csv')
```

```
#Load the dataset and show the first 5 and last 5 datapoints
df.head()
```

	Name	Brand channel	Subscribers (millions)	Primary language	Category	Country
0	MrBeast	No	335	English	Entertainment	United States
1	T-Series	UNKNOWN	280	Hindi	Music	India
2	Cocomelon - Nursery Rhymes	Yes	UNKNOWN	UNKNOWN	Education	United States
3	SET India	Yes	UNKNOWN	Hindi	Entertainment	India
4	Vlad and Niki	No	129	English	Entertainment	Russia

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
#Load the dataset and show the first 5 and last 5 datapoints
df.tail()
```

	Name	Brand channel	Subscribers (millions)	Primary language	Category	Country
45	ARY Digital HD	Yes	59.7	Urdu	Entertainment	Pakistan
46	Fede Vigevani	No	59.4	UNKNOWN	Entertainment	Mexico
47	LooLoo Kids	UNKNOWN	59.3	English	Music	Romania
48	Billion Surprise Toys	Yes	UNKNOWN	English	Entertainment	United States
49	Marshmello	No	57.5	English	Music	United States

```
#Check if there are any missing values in the dataset.
import pandas as pd

df = pd.read_csv('/content/drive/MyDrive/VDA/youtube_subscribers_data.csv')

df.isna().sum()
```

	0
Name	0
Brand channel	0
Subscribers (millions)	0
Primary language	0
Category	0
Country	0

dtype: int64

```
#The df must only have Name, Subscribers, Category, and Country. Remove all other columns.
df = df.drop(columns=['Brand channel', 'Primary language'], errors='ignore')
```

```
df.head()
```

	Name	Subscribers (millions)	Category	Country	
0	MrBeast	335	Entertainment	United States	
1	T-Series	280	Music	India	
2	Cocomelon - Nursery Rhymes	UNKNOWN	Education	United States	
3	SET India	UNKNOWN	Entertainment	India	
4	Vlad and Niki	129	Entertainment	Russia	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
#Convert Subscribers to type int
df['Subscribers (millions)'] = pd.to_numeric(df['Subscribers (millions)'], errors='coerce').astype(int)
```

```
df.head()
```

	Name	Subscribers (millions)	Category	Country	
0	MrBeast	335	Entertainment	United States	
1	T-Series	280	Music	India	
4	Vlad and Niki	129	Entertainment	Russia	
5	Kids Diana Show	128	Entertainment	United States	Ukraine
6	Like Nastya	123	Entertainment	United States	Ukraine

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
#Ensure that there are no rows with missing values. (UNKNOWN is counted as a missing value)
import numpy as np
import pandas as pd

#Creating a .replace(['Condition']), np.nan) to count UNKNOWN as a missing value
df['Category'] = df['Category'].replace(['UNKNOWN', 'unknown', 'Unknown'], np.nan)
df['Country'] = df['Country'].replace(['UNKNOWN', 'unknown', 'Unknown'], np.nan)

#Drop All the rows that contains any NaN / missing values
df = df.dropna()

#displays the df.head() and df.isna().sum()
display(df.head())
display(df.isna().sum())
```

	Name	Subscribers (millions)	Category	Country	
0	MrBeast	335	Entertainment	United States	
1	T-Series	280	Music	India	
4	Vlad and Niki	129	Entertainment	Russia	
5	Kids Diana Show	128	Entertainment	United States	Ukraine
6	Like Nastya	123	Entertainment	United States	Ukraine
		0			
	Name	0			
	Subscribers (millions)	0			
	Category	0			
	Country	0			

dtype: int64

Conclusion

In Conclusion, I've learned how to do data cleaning by the use of Python and Pandas by pre-processing raw data into a structured and reliable dataset for Structural and Visual Analysis. Initial Inspections are showed by `df.isna().sum()` to show the missing values in the dataset. Also, the `df.head()` and `df.tail()` shows the First 5 and Last 5 datapoints on the dataset. Also by adding the `.astype(int)` to this syntax `df['Subscribers (millions)'] = pd.to_numeric(df['Subscribers (millions)'], errors='coerce')`, you can convert the value of a specific column to a Integer Data Type. "pd.to_numeric" is for converting a column into a numeric data type (float64 or int64) and "errors='coerce'" forces any value that cannot be converted into a number to become NaN or Not a Number. By counting UNKNOWN as a missing value from a column, you need to add `.replace(['Condition'], np.nan)` to the `df['Category'] = df['Category']`.

Good job on finishing the activity! Kindly submit your accomplished google colab link and a PDF file of this activity.